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Medical device

Abstract

A medical device includes a liquid circulation unit that enables transfer of a liquid (a perfusion liquid) supplied from a port unit of a second hub of an energy application device to a distal end portion of a first shaft of a catheter device. The energy application device is relatively movable forward and backward with respect to the catheter device so that an energy application unit can be disposed at a predetermined position on the distal end side of the first shaft.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This Application is a 371 of PCT/JP2024/015097 filed on Apr. 16, 2024, which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a medical device.

BACKGROUND ART

(3) There are medical devices known for being used to emit energy in a living organ of a human body to treat and alleviate various diseases and the like. As a treatment method using such a medical device, a procedure for ablating nerves present outside a blood vessel (nerve ablation) is performed.

(4) In a medical device that is used in the above procedure, various kinds of energy can be adopted as the energy to be applied to the treatment target site. As an example of such a medical device, there is a device that adopts the structure of a catheter device that applies thermal energy directly or indirectly to the nerve to be treated from the vicinity of a distal end (see Patent Literature 1, for example).

CITATION LIST

Patent Literature

(5) Patent Literature 1: JP 2017-196461 A

SUMMARY OF INVENTION

Technical Problem

(6) In the catheter device disclosed in Patent Literature 1, energy is emitted from an energy application unit (an ultrasonic radiation source) in a state where the distal end portion of a catheter is disposed at a predetermined position in a blood vessel. When treatment is performed, there is a distance between the energy application unit and the inner wall of the blood vessel, and therefore, it is necessary to radiate energy with high power. Furthermore, when energy is emitted from the energy application unit with high power, the energy application unit and the electric transmission path (a conductive unit) disposed in the catheter may generate heat and have a high temperature. Therefore, the catheter device is required to include a structure for cooling the energy emitting unit with a perfusion liquid.

(7) The present invention has been made in view of the above problems, and aims to provide a medical device that ablates nerves present outside a blood vessel, and is capable of cooling an energy application unit and a conductive unit disposed in a first shaft of a catheter device with a perfusion liquid during treatment.

Solution to Problem

(8) The present invention can be achieved with any one of the following means (1) to (9).

(9) (1)

(10) A medical device including: a catheter device that includes: a first shaft that is configured to be insertable into a blood vessel and has one or more opening formed at a distal end portion; and a first hub disposed at a proximal end portion of the first shaft; an energy application device that includes: an energy application unit configured to apply energy to a surrounding nerve outside the blood vessel, the energy being for ablating the surrounding nerve running in parallel with the blood vessel; a second hub in which an electrical connector unit that supplies electrical energy to the energy application unit is disposed, the second hub including a port unit for inwardly supplying a predetermined liquid; a conductive unit that has an elongated shape and connects the electrical connector unit and the energy application unit; and a second shaft that extends from the second hub to a side of the energy application unit, accommodates at least part of a proximal end side of the conductive unit, and is inserted into the first hub; and a liquid circulation unit that allows the liquid supplied from the port unit to be sent to the distal end portion of the first shaft, in which the energy application device is relatively movable forward and backward with respect to the catheter device,

so that the energy application unit can be placed at a predetermined position on the distal end side of the first shaft.

(2)

(11) The medical device according to (1), in which the liquid circulation unit is a fluid communication passage that includes the port unit of the second hub, an internal space of the second hub, a lumen of the second shaft, an internal space of the first hub, and a lumen of the first shaft.

(12) (3)

(13) The medical device according to (1) or (2), in which the second shaft is inserted into the first hub, to allow a distal end portion of the second shaft to move between a first position defined in the first hub and a second position defined closer to a proximal end side in the first hub than the first position, and the liquid circulation unit is provided in both a state where the distal end portion of the second shaft is located at the first position and a state where the distal end portion of the second shaft is located at the second position.

(4)

(14) The medical device according to any one of (1) to (3), in which the first shaft includes: a first opening formed at a distal end; and a plurality of second openings that is located on a proximal end side of the first opening and is open toward a lateral side of the first shaft.

(5)

(15) The medical device according to (4), in which the plurality of second openings is open in a direction not to face an inner wall of the blood vessel in a state where the first shaft is inserted in the blood vessel.

(16) (6)

(17) The medical device according to (5), in which the distal end portion of the first shaft includes: a curved portion that has a shape curved with respect to a predetermined first reference axis; and a linear portion that is located on a proximal end side of the curved portion, and extends substantially linearly along a second reference axis extending in a direction intersecting with the first reference axis.

(7)

(18) The medical device according to (6), in which the curved portion is a spiral portion that extends spirally around the first reference axis.

(19) (8)

(20) The medical device according to (6) or (7), in which the first shaft includes: a first portion provided at a position including the curved portion; a second portion that is located on a proximal end side of the first portion, and is provided with the plurality of second openings; and a third portion that is located on a proximal end side of the second portion, and is provided at a position including the linear portion, and an inner diameter of the second portion and the third portion is larger than an inner diameter of the first portion.

(9)

(21) The medical device according to any one of (1) to (8), in which the energy application unit is capable of emitting a microwave or an ultrasonic wave as the energy.

Advantageous Effects of Invention

(22) With the medical device according to (1), it is possible to ablate, outside a blood vessel, a surrounding nerve extending in parallel with the blood vessel, by causing energy emission from the energy application unit disposed in the first shaft of the catheter device. Also, the medical device according to (1) includes the liquid circulation unit that enables transfer of the liquid (perfusion liquid) supplied from the port unit of the second hub of the energy application device to the distal end portion of the first shaft. Thus, during treatment, the surgeon can cool the energy application unit and the conductive unit disposed in the first shaft of the catheter device with the perfusion liquid.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram schematically illustrating an overall configuration of a medical device according to an embodiment.
- (2) FIG. 2 is an enlarged cross-sectional view of a portion indicated as a dashed-line portion 2A in FIG. 1.
- (3) FIG. 3 is an enlarged cross-sectional view of a portion indicated as a dashed-line portion 3A in FIG. 1.
- (4) FIG. 4 is an enlarged cross-sectional view of a portion indicated as a dashed-line portion 4A in FIG. 1.
- (5) FIG. 5 is an enlarged perspective view of a portion of a first shaft.
- (6) FIG. 6 is an enlarged perspective view of the vicinity of a distal end portion of the first shaft.
- (7) FIG. 7 is a partial cross-sectional view of the vicinity of the distal end portion of the first shaft.
- (8) FIG. 8 is a cross-sectional view (a transverse sectional view) of a portion indicated by arrows 8A-8A in FIG. 6.
- (9) FIG. 9 is a cross-sectional view (a transverse sectional view) of a portion indicated by arrows 9A-9A in FIG. 6.
- (10) FIG. 10 is a cross-sectional view of an inserter included in the medical device.
- (11) FIG. 11 is a view for explaining an example of use of the inserter included in the medical device.
- (12) FIG. 12 is a schematic cross-sectional view (a longitudinal sectional view in the extending direction of a blood vessel) for explaining an example of use of the medical device.
- (13) FIG. 13 is a cross-sectional view (a transverse sectional view) of a portion indicated by arrows 13A-13A in FIG. 12.

DESCRIPTION OF EMBODIMENTS

- (14) In the description below, embodiments of the present invention will be explained with reference to the accompanying drawings. Note that the following description does not limit the technical scope of the claims or the meanings of terms used in the claims. Also, dimensional ratios in the drawings are exaggerated for ease of explanation, and may differ from actual ratios. Further, a numerical range “X to Y” in the present specification means “not smaller than X and not greater than Y”.
- (15) FIGS. 1 to 11 are diagrams for explaining the respective components of a medical device 10 according to an embodiment. FIGS. 12 and 13 are diagrams for explaining examples of use of the medical device 10. Note that FIG. 12 is a longitudinal sectional view in the extending direction of a blood vessel V to be treated, and FIG. 13 is a transverse sectional view of the blood vessel V to be treated (a transverse sectional view of a portion indicated by an arrow 13A-13A in FIG. 12).
- (16) As illustrated in FIGS. 12 and 13, in a procedure using the medical device 10, the surgeon (a medical worker such as a medical doctor) can enhance peristaltic movement of the intestinal tract by performing a treatment for reducing the activity of the autonomic nerves in the blood vessel V having surrounding nerves (extravascular nerves) Na that innervate the intestinal tract of the patient. By performing such a treatment, the surgeon can facilitate alleviation of at least one symptom among abdominal distension, abdominal pain, perineal discomfort, and frequent defecation caused by constipation and/or an abnormality in peristaltic movement of the intestinal tract of the patient (at least one symptom among the symptom groups caused by alleviation of constipation and/or an abnormality in peristaltic movement of the intestinal tract of the patient).
- (17) The blood vessel V to which the treatment method is applied is not limited to any particular kind, as long as the peristaltic movement of the intestinal tract of the patient can be enhanced by a predetermined treatment (denervation by application of energy, for example) according to the

embodiment. As an example of the blood vessel V, at least one blood vessel can be suitably selected from among the superior mesenteric artery Va, the celiac artery, and the inferior mesenteric artery. In particular, the superior mesenteric artery Va can be selected as the blood vessel V from the viewpoint of more effectively enhancing the peristaltic movement of the intestinal tract after the treatment. In the present embodiment, an example in which treatment is performed on the superior mesenteric artery Va will be described (see FIGS. 12 and 13).

(18) As the treatment according to the embodiment, the surgeon applies energy to one surrounding nerve Na (or a plurality of surrounding nerves) that is present outside the blood vessel V and extends in parallel with the blood vessel V. By applying energy to the surrounding nerve Na and impairing the surrounding nerve Na, the surgeon completely or partially blocks autonomic nerve transmission to the digestive tract from the surrounding nerve Na. By performing such treatment, the surgeon can enhance peristaltic movement of the intestinal tract.

(19) The following mechanisms are considered as the reasons why peristaltic movement of the intestinal tract is activated by performing the above treatment.

(20) When the surrounding nerve Na is impaired by the energy applied through the inside of the blood vessel V, and the autonomic nerve transmission to the digestive tract by the surrounding nerve Na is completely or partially blocked, the sympathetic nervous system weakens relatively and the parasympathetic nervous system becomes dominant between the sympathetic nervous system and the parasympathetic nervous system. Also, when neurotransmission from the central nerves is blocked, the intestinal nervous system that autonomously controls intestinal movement in the periphery becomes dominant, and peristaltic movement of the intestinal tract becomes active. Further, as peristaltic movement of the intestinal tract becomes active, the colon transit time is accelerated and normalized, and thus, alleviation of at least one symptom among abdominal distension, abdominal pain, perineal discomfort, and frequent defecation caused by constipation and/or an abnormality in peristaltic movement of the intestinal tract is facilitated.

(21) In particular, by the treatment method according to the present embodiment, it is possible to suitably accelerate alleviation of a symptom of constipation of a colon transit time delay type that is derived from degradation in intestinal peristaltic movement of the large intestine and a delay observed in the stool transit time in functional constipation without an organic abnormality in the colon.

(22) The treatment target site (a region including one or a plurality of surrounding nerves Na illustrated in FIG. 13) S of the treatment using the medical device 10 is not limited to any particular site, as long as it is possible to enhance peristaltic movement of the intestinal tract. For example, in the blood vessel V, the treatment may be performed on any appropriate range (site) in the running direction (extending direction) of the blood vessel V, or the treatment may be performed on any appropriate range (site) in a circumferential direction (a circumferential direction of the transverse cross-section) of the blood vessel V. Also, the treatment may be performed a plurality of times at a plurality of appropriate portions of the same blood vessel V, or may be performed a plurality of times at any appropriate portions of a plurality of different blood vessels V.

(23) In the present embodiment, a treatment method that regards the superior mesenteric artery Va as the blood vessel V to be treated as described above will be described.

(24) The treatment method includes performing treatment on a site around the origin of the superior mesenteric artery Va. Specifically, the treatment target site S preferably includes a range of 0 mm to 20 mm in the extending direction of the superior mesenteric artery Va, with reference to the opening of the superior mesenteric artery Va (an opening connected to the aorta), for example. By applying energy within the above range of the superior mesenteric artery Va, it is possible to effectively suppress energy transmission to organs (such as the pancreas or the duodenum, for example) located on the peripheral side of the superior mesenteric artery Va.

(25) Also, from the viewpoint of more reliably suppressing energy transmission to the organs located on the peripheral side of the superior mesenteric artery Va, it is still more preferable to

apply energy from the superior mesenteric artery Va only within the range of 0 mm to 20 mm in the extending direction of the superior mesenteric artery Va, for example.

(26) The invasion depth of the energy from an energy application unit **210** disposed in the superior mesenteric artery Va toward the outside of the blood vessel V preferably includes a range of 1 mm to 6 mm from the intima of the superior mesenteric artery Va, for example. The surrounding nerves Na present outside the superior mesenteric artery Va are present at relatively deep positions around the origin of the superior mesenteric artery Va. More specifically, the surrounding nerves Na exist in a bundle while being supported by connective tissue in adipose tissue outside the superior mesenteric artery Va. Accordingly, in a case where energy is applied from around the origin of the superior mesenteric artery Va, the surrounding nerves Na around the superior mesenteric artery Va can be efficiently denervated by causing the energy to reach a position of 1 mm to 6 mm of the intima of the superior mesenteric artery Va.

(27) Next, the medical device **10** according to the present embodiment is described.

(28) <Medical Device **10**>

(29) As illustrated in FIGS. **1** to **4**, the medical device **10** includes a catheter device **100**, an energy application device **200**, and a liquid circulation unit **50**.

(30) As illustrated in FIGS. **1** to **4**, **12**, and **13**, the catheter device **100** includes a first shaft **110** that is configured to be insertable into the blood vessel V and has one or more openings **112a** and **112b** formed at a distal end portion **111**, and a first hub **120** disposed at a proximal end portion **113** of the first shaft **110**.

(31) Also, the energy application device **200** includes: the energy application unit **210** that is configured to be able to apply energy for ablating the surrounding nerve Na that is located outside the blood vessel V and extends in parallel with the blood vessel V; a second hub **220** in which an electrical connector unit **240** for supplying electrical energy to the energy application unit **210** is disposed, and a port unit **224** for supplying a predetermined liquid to the inside is included; an elongated conductive unit **260** that connects the electrical connector unit **240** and the energy application unit **210**; and a second shaft **230** that extends from the second hub **220** toward the energy application unit **210**, accommodates at least part of the proximal end side of the conductive unit **260**, and is inserted into the first hub **120**.

(32) In the description of the present specification, the side of the first shaft **110** of the catheter device **100** to be inserted into the blood vessel V (the side indicated by an arrow X1) is defined as the distal end side, and the side of the energy application device **200** that is located on the opposite side to the distal end side and is the side on which the second hub **220** is disposed (the side indicated by an arrow X2) is defined as the proximal end side.

(33) <Catheter Device **100**>

(34) As illustrated in FIGS. **1**, **2**, and **7**, the first shaft **110** included in the catheter device **100** can be formed with a tubular member having a lumen **115** formed therein.

(35) As illustrated in FIGS. **5**, **6**, and **7**, the first shaft **110** has a first opening **112a** formed at the distal end, and a plurality of second openings (side holes) **112b** that is located on the proximal end side of the first opening **112a** and is open toward the lateral side of the first shaft **110**.

(36) The catheter device **100** can supply a contrast medium to the lumen **115** of the first shaft **110** via a port unit **124** of the first hub **120**. The contrast medium supplied to the lumen **115** of the first shaft **110** can be discharged to the outside of the first shaft **110** through the plurality of second openings **112b** and the first opening **112a**.

(37) The energy application device **200** can supply liquid (saline, for example) to a lumen **235** of the second shaft **230** via the port unit **224** of the second hub **220**. As described later, the lumen **235** of the second shaft **230** communicates with the lumen **115** of the first shaft **110** via the first hub **120**. Therefore, the liquid supplied via the port unit **224** of the second hub **220** can be sent into the lumen **115** of the first shaft **110**, and be discharged to the outside of the first shaft **110** via the plurality of second openings **112b** and the first opening **112a** of the first shaft **110**.

(38) As illustrated in FIGS. 6 and 7, the first shaft **110** includes a distal end tip **140** formed with a flexible resin material. The distal end tip **140** has the first opening **112a** that communicates with the lumen **115** of the first shaft **110**. The first opening **112a** forms a distal end opening that faces the distal end side of the first shaft **110**.

(39) Note that the first shaft **110** does not necessarily include the distal end tip **140**. In a case where the first shaft **110** does not include the distal end tip **140**, the first opening **112a** can be formed with an opening provided at the distal end of the first shaft **110**.

(40) As illustrated in FIG. 6, the distal end portion **111** of the first shaft **110** includes: a curved portion **116** having a curved shape with respect to a predetermined first reference axis **A1**; and a linear portion **118** that is located on the proximal end side of the curved portion **116** and extends substantially linearly along a second reference axis **A2** extending in a direction intersecting with the first reference axis **A1**.

(41) As the first shaft **110** includes the curved portion **116**, at least part of the distal end portion **111** of the first shaft **110** can be disposed so as to come into contact with an inner wall **Vi** of the blood vessel **V** when the distal end portion **111** of the first shaft **110** is inserted into the blood vessel **V** (see FIGS. 12 and 13). In a state where at least part of the distal end portion **111** of the first shaft **110** is in contact with the inner wall **Vi** of the blood vessel **V**, the surgeon causes the energy application unit **210** disposed in the lumen **115** of the distal end portion **111** of the first shaft **110** to emit energy, so that energy can be efficiently transmitted to the surrounding nerve **Na** located outside the blood vessel **V**.

(42) Further, as the first shaft **110** has the linear portion **118** disposed on the proximal end side with respect to the curved portion **116**, the first shaft **110** can be disposed so that a position on the proximal end side with respect to the curved portion **116** comes into contact with the upper end of the blood vessel **V** (the upper end around the origin from the aorta to the superior mesenteric artery **Va**, for example) when the first shaft **110** is inserted into the blood vessel **V**. By placing the linear portion **118** of the first shaft **110** so as to come into contact with the inner wall **Vi** of the blood vessel **V** of the blood vessel **V**, the surgeon can cause the linear portion **118** to firmly support the first shaft **110** with respect to the blood vessel **V**, and prevent the curved portion **116** from being displaced.

(43) As illustrated in FIG. 6, the curved portion **116** can be formed with a spiral portion **117** extending spirally around the first reference axis **A1**, for example.

(44) In the catheter device **100**, the curved portion **116** is formed with the spiral portion **117**. Therefore, when the first shaft **110** is inserted into the blood vessel **V**, the spiral portion **117** extends spirally in the blood vessel **V**, and comes into contact with the inner wall **Vi** of the blood vessel **V** in a continuous state in a predetermined range. Accordingly, as illustrated in FIG. 12, the portion at which the spiral portion **117** is formed can be continuously brought into contact with the inner wall **Vi** of the blood vessel **V**. As a result, in the catheter device **100**, the energy application unit **210** can be disposed at any desired portion in the extending direction of the blood vessel **V** and the circumferential direction of the blood vessel **V** within the range in which the spiral portion **117** is formed. Thus, energy can be easily and efficiently applied to the surrounding nerve **Na** from a position close to the inner wall **Vi** of the blood vessel **V**.

(45) A specific shape (such as the pitch of the spiral, the number of turns, the outer diameter of the spiral, and the winding direction of the spiral, for example) of the spiral portion **117** is not limited to any particular shape, as long as any desired portion of the spiral portion **117** can be brought into contact with the inner wall **Vi** of the blood vessel **V** when the first shaft **110** is disposed in the blood vessel **V**.

(46) Also, the curved portion **116** is not limited only to a shape like that of the spiral portion **117**, as long as at least part of the distal end portion **111** of the first shaft **110** can be brought into contact with the inner wall **Vi** of the blood vessel **V**. The curved portion **116** can also be configured to have a shape curved in a zigzag form with respect to the first reference axis **A1**, a circular shape

intersecting with the first reference axis **A1**, an elliptical shape, a rectangular shape, a U-shape folded back from the distal end side to the proximal end side, or the like, for example.

(47) As illustrated in FIG. **12**, the plurality of second openings **112b** is open in a direction not to face the inner wall **Vi** of the blood vessel **V** in a state where the first shaft **110** is inserted in the blood vessel **V**. Note that the direction in which the plurality of second openings **112b** are open can be defined as a direction in which the first reference axis **A1** serving as a reference for the spiral shape of the spiral portion **117** is oriented inward (see FIG. **6**).

(48) The plurality of second openings **112b** is located at a second portion **119b** on the proximal end side of the spiral portion **117**. In the present embodiment, each of the second openings **112b** is formed with a circular hole. Further, three second openings **112b** are arranged at predetermined intervals so as to be disposed at different positions in the extending direction of the first shaft **110**. The opening area of each of the second openings **112b** is smaller than the opening area of the first opening **112a**.

(49) Since each opening of the plurality of second openings **112b** is open in a direction not to face the inner wall **Vi** of the blood vessel **V**, it is possible to prevent the second openings **112b** from being blocked by the inner wall **Vi** of the blood vessel **V** when the first shaft **110** is inserted into the blood vessel **V**. Furthermore, the plurality of second openings **112b** is disposed at predetermined positions on the proximal end side away from the first opening **112a** located at the distal end of the first shaft **110**. Accordingly, even when the distal end side of the first shaft **110** is inserted to a portion at which the cross-sectional area on the back side of the blood vessel **V** is narrowed, it is possible to image the portion on the proximal side of the blood vessel **V**, by causing each of the second openings **112b** to discharge the contrast medium.

(50) Note that the number, the placement locations, the shape, the opening area, and the like of the second openings **112b** are not limited to any particular kinds, as long as the contrast medium can be discharged from a position on the proximal end side of the first opening **112a**.

(51) As illustrated in FIGS. **5**, **6**, and **7**, the first shaft **110** includes: a first portion **119a** provided at a position including the curved portion **116**; the second portion **119b** that is located on the proximal end side of the first portion **119a**, and has the plurality of second openings **112b** formed therein; and a third portion **119c** that is located on the proximal end side of the second portion **119b**, and is provided at a position including the linear portion **118**.

(52) As illustrated in FIGS. **8** and **9**, the inner diameter **d2** of the second portion **119b** and the third portion **119c** can be configured to be larger than the inner diameter **d1** of the first portion **119a**. Although FIG. **9** shows an axis-orthogonal cross-sectional view of the second portion **119b**, the third portion **119c** is also configured to have a cross-sectional shape (with the inner diameter **d2** or the like) substantially similar to that of the second portion **119b** in the present embodiment.

(53) In the first shaft **110**, the inner diameter of the first portion **119a** provided in a predetermined range from the distal end to the proximal end side of the first shaft **110** is relatively small. Thus, it is possible to prevent the energy application unit **210** from being disposed at a position shifted radially outward from the center position on an axis-orthogonal cross-section of the lumen **115** at a portion corresponding to the first portion **119a** of the lumen **115**. For example, if the energy application unit **210** is disposed at a position largely shifted radially outward from the center position as described above at a time of treatment, there is a possibility that a deviation in intensity distribution of the energy emitted from the energy application unit **210** or a deviation in resonance (in a case where the energy application unit **210** is formed with an antenna element) will occur. With the first shaft **110** including the first portion **119a**, it is possible to prevent the occurrence of such a problem in advance.

(54) Further, in the first shaft **110**, the inner diameters of the second portion **119b** and the third portion **119c** located on the proximal end side of the first portion **119a** are relatively large. Thus, it is possible to ensure large clearances for the lumen **115** at the portions corresponding to the respective portions **119b** and **119c**. Accordingly, flows of the contrast medium and the perfusion

liquid in the respective portions **119b** and **119c** can be made smooth, and the contrast medium and the perfusion liquid can be more reliably discharged from the second openings **112b** formed in the second portion **119b**.

(55) As illustrated in FIGS. **8** and **9**, the portion of the first shaft **110** corresponding to the first portion **119a** can be configured to have a tube wall thickness smaller than that of the portions of the first shaft **110** corresponding to the second portion **119b** and the third portion **119c**. Thus, in the first shaft **110**, the inner diameter **d2** of each of the portions **119b** and **119c** can be configured to be larger than the inner diameter **d1** of the first portion **119a**.

(56) Note that the first portion **119a** can be configured to have a higher flexibility than the second portion **119b** and the third portion **119c**. With such a configuration, in the first shaft **110**, the flexibility of the first portion **119a** in which the curved portion **116** is formed is high, and the rigidity of each of the portions **119b** and **119c** located on the proximal end side of the first portion **119a** is high. Accordingly, the first shaft **110** having a high supporting force capable of supporting the contact of the curved portion **116** with the inner wall **Vi** of the blood vessel **V** at each of the portions **119b** and **119c** on the proximal end side of the curved portion **116** while preventing the inner wall **Vi** of the blood vessel **V** from being damaged by the curved portion **116** configured so that at least a portion thereof comes into contact with the inner wall **Vi** of the blood vessel **V** is formed. The magnitude relationship in terms of flexibility can be adjusted with the types of the materials forming the respective portions **119a**, **119b**, and **119c** of the first shaft **110**, for example.

(57) As illustrated in FIG. **6**, part of the distal end side of the second portion **119b** can be configured to have a predetermined curved shape between the linear portion **118** and the curved portion **116**. Also, this portion can be configured to have a curved shape with a larger curvature radius than that of the spiral portion **117**, for example. As the second portion **119b** has a portion with a relatively large curvature radius as described above, it is possible to prevent the flows of the contrast medium and the perfusion liquid from being stagnant in the second portion **119b**. Thus, the contrast medium and the perfusion liquid can be more smoothly discharged from the second openings **112b** formed in the second portion **119b**.

(58) Note that the portion of the first shaft **110** extending to the proximal end side of the third portion **119c** can be configured to have substantially the same cross-sectional shape (the inner diameter or the like) as the third portion **119c**.

(59) The first shaft **110** can be formed with a resin material that is known in the catheter field, for example. Also, the first shaft **110** may be formed with a single-layer or multilayer resin tube member, or may have a blade wire or the like embedded therein to reinforce its rigidity. Further, the outer surface of the first shaft **110** may be coated with a hydrophilic coating or the like. These aspects also apply to the second shaft **230** described later.

(60) As illustrated in FIGS. **2** and **3**, the first hub **120** includes: a distal cylindrical portion **121** that is located at the distal end portion; a hub main body **123** in which an internal space **125** communicating with the inside of the distal cylindrical portion **121** is provided; the port unit **124** that has a flow path communicating with the internal space **125** of the hub main body **123**; and an opener **127** that is disposed on the proximal end side of the port unit **124**.

(61) The first hub **120** can be formed with a Y connector (a Y connector or the like with a valve mechanism that can be opened and closed by the opener **127**, for example) that is known in the catheter field, for example.

(62) A wing **130** can be disposed on the distal end side of the distal cylindrical portion **121** of the first hub **120**. As the wing **130**, one that is known in the catheter field can also be used like the Y connector. The proximal end portion **113** of the first shaft **110** is secured to the wing **130**.

(63) A proximal end opening **113a** provided at the proximal end portion **113** of the first shaft **110** is disposed so as to communicate with an internal space **135** of the wing **130**. The internal space **135** of the wing **130** communicates with the internal space **125** of the hub main body **123** via the distal cylindrical portion **121**. Accordingly, when the contrast medium is supplied from the port unit **124**

of the first hub **120**, the contrast medium flows into the lumen **115** of the first shaft **110** via the port unit **124**, the internal space **125** of the hub main body **123**, the internal space **135** of the wing **130**, and the proximal end opening **113a**. The contrast medium that has flowed into the lumen **115** of the first shaft **110** is discharged to the outside of the first shaft **110** via the first opening **112a** provided at the distal end portion **111** of the first shaft **110** and the plurality of second openings **112b**.

(64) As illustrated in FIG. 3, the opener **127** is disposed at a position on the proximal end side of a proximal end portion **126** of the hub main body **123**. The second shaft **230** included in the energy application device **200** is inserted through the inside of the opener **127** and is inserted into the internal space **125** of the hub main body **123**.

(65) As illustrated in FIGS. 1, 10, and 11, the medical device **10** can include a predetermined inserter **300**.

(66) The inserter **300** includes: a distal end opening **301a** provided at a distal end portion **301**; a proximal end opening **303a** provided at a proximal end portion **303**; and a lumen **305** that communicates with the distal end opening **301a** and the proximal end opening **303b**.

(67) A tapered portion whose outer diameter and inner diameter become gradually smaller in the direction toward the distal end side is provided at the distal end portion **301** of the inserter **300**. Also, a tapered portion whose outer diameter and inner diameter become gradually larger in the direction toward the proximal end side is provided at a position on the distal end side of the proximal end portion **303** of the inserter **300**.

(68) At the start of a procedure using the medical device **10**, the inserter **300** is disposed near the distal end portion **111** of the first shaft **110** as illustrated in FIG. 10. As the inserter **300** is disposed near the distal end portion **111** of the first shaft **110**, the shaped curved portion **116** (the spiral portion **117**) of the first shaft **110** is straightened into a substantially linear shape along the inner surface shape of the inserter **300**.

(69) Before inserting the medical device **10** into the blood vessel V, the surgeon removes the energy application device **200** from the catheter device **100**. In the state where the energy application device **200** has been removed from the catheter device **100**, the surgeon uses the inserter **300** to correct the curved portion **116** of the first shaft **110** to have a substantially linear shape as described above. In the state where the curved portion **116** has been corrected to have a substantially linear shape, the surgeon inserts a guide wire over the entire length direction of the catheter device **100** via the first opening **112a** of the first shaft **110**, starting from the proximal end side of the guide wire inserted into the blood vessel V prior to the medical device **10**. By moving the catheter device **100** along the guide wire, the surgeon can smoothly insert a certain range of the distal end portion **111** of the first shaft **110** of the catheter device **100** up to a predetermined position in the blood vessel V.

(70) After inserting the certain range of the distal end portion **111** of the first shaft **110** into the blood vessel V in the above procedure, the surgeon removes the guide wire from the living body via the catheter device **100**. After used for insertion of the medical device **10**, the inserter **300** is moved along the outer surface of the first shaft **110** to the vicinity of the first hub **120**, as illustrated in FIGS. 1 and 11. The inserter **300** can be configured to be connectable to the first hub **120**, for example. The inserter **300** can be used as an anti-kink protector that protects the first shaft **110** by partially covering the first shaft **110** in a state of being connected to the first hub **120**.

(71) After removing the guide wire from the living body via the catheter device **100**, the surgeon inserts the energy application device **200** from the first hub **120** of the catheter device **100**. By moving the energy application device **200** along the first shaft **110** of the catheter device **100**, the surgeon can place the energy application unit **210** at a predetermined position in the distal end portion **111** of the first shaft **110**. Note that, using the opener **127**, the surgeon can seal the portion between the first hub **120** and the second shaft **230** of the energy application device **200**, to prevent leakage of liquid or the like to the proximal end side of the first hub **120**.

(72) The first shaft **110** has a structure that is shaped in advance so that the curved portion **116** (the

spiral portion **117**) is formed in at least part of the distal end portion **111** in a natural state where no external force is applied thereto. Accordingly, when the guide wire is removed from the first shaft **110**, and the inserter **300** is moved to the proximal end side of the distal end portion **111**, the curved portion **116** (the spiral portion **117**) is restored to a predetermined curved shape, and at least a portion thereof is brought into contact with the inner wall V_i of the blood vessel V (see FIGS. **12** and **13**).

(73) <Energy Application Device **200**>

(74) As illustrated in FIG. **13**, the energy application unit **210** is configured to be capable of emitting energy that can reach the surrounding nerve Na located outside the blood vessel V , in a state of being disposed in the lumen **115** of the first shaft **110**.

(75) The energy application unit **210** can be formed with an antenna element capable of emitting microwaves, for example. In a case where the energy application unit **210** is formed with an antenna element, the center frequency of the antenna element can be set to any of 915 MHz, 2.45 GHz, 5.8 GHz, and 24.125 GHz, for example.

(76) The energy application unit **210** can also be formed with an ultrasonic element capable of emitting ultrasonic waves, for example. However, the energy application unit **210** is not limited to any particular structure, as long as the surrounding nerve Na extending in parallel with the outside the blood vessel V can be ablated. The energy application unit **210** can have a configuration capable of ablating the surrounding nerve Na , using a simple high frequency, a bipolar high frequency, high-density focusing ultrasonic waves, light, heat, cold radiation, an engineering therapy, a magnetic, electrical, or electromagnetic cryotherapy, plasma, mechanical energy, chemical energy, kinetic energy, potential energy, nuclear energy, or the like, for example.

(77) As illustrated in FIGS. **1** and **4**, the second hub **220** of the energy application device **200** is disposed on the proximal end side of the first hub **120** of the catheter device **100**.

(78) The electrical connector unit **240** is accommodated in an internal space **225** of the second hub **220**. The electrical connector unit **240** is covered with a predetermined housing **250**.

(79) The electrical connector unit **240** is configured to be connectable to a power supply unit **400**, which is an external device, via a cord **410**. Various electrical elements and the like for receiving power from the power supply unit **400** are disposed in the electrical connector unit **240**.

(80) The energy application unit **210** and the electrical connector unit **240** are connected via the conductive unit **260**. The energy application unit **210** receives power from the electrical connector unit **240** via the conductive unit **260**.

(81) In a case where the energy application unit **210** is formed with an antenna element, the conductive unit **260** can be formed with a known coaxial cable capable of supplying current to the antenna element. Note that, in a case where the energy application unit **210** is formed with a component other than an antenna element, the conductive unit **260** can be formed with various electrical structural members that enable transmission of energy to the energy application unit **210**.

(82) As illustrated in FIGS. **1** and **4**, an anti-kink protector **227** can be disposed at the proximal end portion of the second hub **220** of the energy application device **200**.

(83) As illustrated in FIG. **4**, the internal space **225** of the second hub **220** communicates with the port unit **224**. A clearance through which the perfusion liquid supplied from the port unit **224** can flow is provided between the electrical connector unit **240** accommodated in the internal space **225** of the second hub **220** and the inner surface of the second hub **220**.

(84) Note that, in the internal space **225** of the second hub **220**, a portion closer to the proximal end side than the vicinity of the portion communicating with the port unit **224** is closed so that the perfusion liquid does not move closer to the proximal end side than the port unit **224**.

(85) A proximal end portion **233** of the second shaft **230** is connected to the distal end portion of the second hub **220**. The internal space **225** of the second hub **220** communicates with a proximal end opening **233a** provided in the proximal end portion **233** of the second shaft **230**.

(86) As illustrated in FIG. **2**, the second shaft **230** extends from the second hub **220** to the distal end

side, and its distal end portion **231** is disposed in the internal space **125** of the first hub **120**.

(87) The conductive unit **260** is inserted into the lumen **115** of the first shaft **110** via the internal space **225** of the second hub **220**, the second shaft **230**, the internal space **125** of the first hub **120**, and the wing **130**.

(88) In the energy application device **200**, the second shaft **230** can relatively move forward and backward with respect to the catheter device **100** so that the energy application unit **210** can be disposed at a predetermined position (a predetermined position of the curved portion **116**, for example) on the distal end side of the first shaft **110**.

(89) An arrow *f* in FIGS. **1**, **2**, and **7** indicates a direction in which the energy application unit **210** and part of the second shaft **230** move forward along the lumen **115** of the first shaft **110**. An arrow *b* indicates a direction in which the energy application unit **210** and part of the second shaft **230** move backward along the lumen **115** of the first shaft **110**.

(90) The surgeon grips the second hub **220** with fingers or the like, and causes the second hub **220** to slide forward and backward along a central axis *C1* of the second shaft **230**, so that the energy application device **200**, together with the second hub **220**, can relatively move forward and backward with respect to the catheter device **100**.

(91) The surgeon can selectively place the energy application unit **210** at any desired portion of the curved portion **116**, by moving the energy application unit **210** along the curved portion **116** in a state where part of the curved portion **116** (the spiral portion **117**) is in contact with the inner wall *Vi* of the blood vessel *V* as illustrated in FIGS. **12** and **13**.

(92) <Liquid Circulation Unit **50**>

(93) The liquid circulation unit **50** enables transfer of the perfusion liquid supplied from the port unit **224** of the second hub **220** to the distal end portion **111** of the first shaft **110**.

(94) Having the liquid circulation unit **50**, the medical device **10** can cool the electrical connector unit **240** disposed in the internal space **225** of the second hub **220**, the conductive unit **260** extending from the electrical connector unit **240** to the vicinity of the distal end portion **111** of the first shaft **110**, and the energy application unit **210** disposed at the distal end of the conductive unit **260**, with the perfusion liquid circulated via the liquid circulation unit **50**. Accordingly, it is possible to prevent the occurrence of an excessive temperature rise in each of the units **240**, **260**, and **210** during treatment using the medical device **10**.

(95) As illustrated in FIGS. **2**, **3**, and **4**, the liquid circulation unit **50** can be formed with a fluid communication passage that includes the port unit **224** of the second hub **220**, the internal space **225** of the second hub **220**, the lumen **235** of the second shaft **230**, the internal space **125** of the first hub **120**, and the lumen **115** of the first shaft **110**. That is, the perfusion liquid supplied via the port unit **224** of the second hub **220** flows from the proximal end side to the distal end side in each of the portions **225**, **235**, **125**, and **115**, to cool the electrical connector unit **240**, the conductive unit **260**, and the energy application unit **210** in the course of flowing. The perfusion liquid that has flowed to the distal end portion **111** of the first shaft **110** is discharged to the outside of the first shaft **110** via the plurality of second openings **112b** and the first opening **112a** formed in the first shaft **110**.

(96) As illustrated in FIG. **2**, the second shaft **230** is inserted into the first hub **120** so that the distal end portion **231** of the second shaft **230** becomes movable between a first position *P1* defined in the first hub **120** and a second position *P2* defined closer to the proximal end side in the first hub **120** than the first position *P1*. Note that specific positions of the first position *P1* and the second position *P2* are not limited to any particular positions, and can be set at any desired positions in the first hub **120**.

(97) In the medical device **10**, the liquid circulation unit **50** is provided in both a state where the distal end portion **231** of the second shaft **230** is located at the first position *P1* and a state where the distal end portion **231** is located at the second position *P2*.

(98) Specifically, when the distal end portion **231** of the second shaft **230** is located at the first

position P1, the distal end portion **231** of the second shaft **230** is in contact with the portion protruding toward the inner peripheral side of the distal cylindrical portion **121**, and the first position P1 of the distal end portion **231** of the second shaft **230** is defined. In this state, the perfusion liquid can flow to the distal end side of the distal end portion **231** of the second shaft **230** via a distal end opening **231a** of the second shaft **230**. On the other hand, when the distal end portion **231** of the second shaft **230** is located at the second position P2, the perfusion liquid can flow toward the distal end side of the distal end portion **231** of the second shaft **230** via the distal end opening **231a** of the second shaft **230** and the internal space **125** of the hub main body **123**.

(99) As described above, the medical device **10** according to the present embodiment includes: the catheter device **100** that includes: the first shaft **110** that is configured to be insertable into the blood vessel V and has one or more openings **112a** and **112b** formed at the distal end portion **111**; and the first hub **120** disposed at the proximal end portion **113** of the first shaft **110**; the energy application device **200** that includes: the energy application unit **210** that is configured to be capable of applying energy to the surrounding nerve Na extending in parallel with the blood vessel V, to ablate the surrounding nerve Na outside the blood vessel V; the second hub **220** in which the electrical connector unit **240** for supplying electrical energy to the energy application unit **210** is disposed, the second hub **220** including the port unit **224** that supplies a predetermined liquid to the inside of the second hub **220**; the elongated conductive unit **260** that connects the electrical connector unit **240** and the energy application unit **210**; and the second shaft **230** that extends from the second hub **220** toward the energy application unit **210**, accommodates at least part of the proximal end side of the conductive unit **260**, and is inserted into the first hub **120**; and the liquid circulation unit **50** that enables transfer of the liquid supplied from the port unit **224** to the distal end portion **111** of the first shaft **110**, in which the energy application device **200** is capable of relatively moving forward and backward with respect to the catheter device **100** so that the energy application unit **210** can be disposed at a predetermined position on the distal end side of the first shaft **110**.

(100) With the medical device **10**, it is possible to ablate, outside the blood vessel V, the surrounding nerve Na extending in parallel with the blood vessel V, by causing energy emission from the energy application unit **210** disposed in the first shaft **110** of the catheter device **100**. Also, the medical device **10** includes the liquid circulation unit **50** that enables transfer of the liquid (perfusion liquid) supplied from the port unit **224** of the second hub **220** of the energy application device **200** to the distal end portion **111** of the first shaft **110**. Thus, during treatment, the surgeon can cool the energy application unit **210** and the conductive unit **260** disposed in the first shaft **110** of the catheter device **100** with the perfusion liquid.

(101) Although a medical device according to the present invention has been described so far through an embodiment, the present invention is not limited only to the details described in the specification, and can be appropriately changed on the basis of the disclosure in the claims.

(102) In the above-described embodiment, the superior mesenteric artery Va has been described as an example of the blood vessel V to be treated by the medical device **10**. However, the blood vessel V to be treated by the medical device is not limited to any particular kind. For example, it is possible to select a pulmonary vein as the blood vessel to be treated. By locally applying energy to the surrounding nerves running outside the pulmonary artery and performing ablation, it is possible to treat a disease such as atrial fibrillation and facilitate recovery.

(103) Also, in the medical device, any component or the like not specifically described in the specification can be added as appropriate, and the additional members described in the specification can be omitted as appropriate. Further, as for the treatment using the medical device, any procedure not specifically described in the specification can be added as appropriate, and the additional procedures described in the specification can be omitted as appropriate. Furthermore, in the treatment method, the sequence of procedures can be changed as appropriate, as long as the effects of the invention can be achieved.

REFERENCE SIGNS LIST

(104) **10** Medical device **50** Liquid circulation unit **100** Catheter device **110** First shaft **111** Distal end portion of the first shaft **112a** First opening **112b** Second opening **113** Proximal end portion of the first shaft **113a** Proximal end opening of the first shaft **115** Lumen of the first shaft **116** Curved portion **117** Spiral portion **118** Linear portion **119a** First portion **119b** Second portion **119c** Third portion **120** First hub **121** Distal cylindrical portion **123** Hub main body **124** Port unit of the first hub **125** Internal space of the hub main body **200** Energy application device **210** Energy application unit **220** Second hub **224** Port unit of the second hub **225** Internal space of the second hub **230** Second shaft **231** Distal end portion of the second shaft **231a** Distal end opening of the second shaft **233** Proximal end portion of the second shaft **233a** Proximal end opening of the second shaft **235** Lumen of the second shaft **240** Electrical connector unit **260** Conductive unit **300** Inserter **400** Power supply unit **A1** First reference axis **A2** Second reference axis **C1** Central axis of the second shaft **Na** Surrounding nerve **OA** Center position of a blood vessel **P1** First position **P2** Second position **S** Treatment target site **V** Blood vessel **Vi** Inner wall of the blood vessel **Va** Superior mesenteric artery

Claims

1. A medical device comprising: a catheter device that includes: a first shaft that is configured to be insertable into a blood vessel and has one or more openings formed at a distal end portion; and a first hub disposed at a proximal end portion of the first shaft; an energy application device that includes: an energy application unit configured to apply energy to a surrounding nerve outside the blood vessel, the energy being for ablating the surrounding nerve running in parallel with the blood vessel; a second hub in which an electrical connector unit that supplies electrical energy to the energy application unit is disposed, the second hub including a port unit for inwardly supplying a predetermined liquid; a conductive unit that has an elongated shape and connects the electrical connector unit and the energy application unit; and a second shaft that extends from the second hub to a side of the energy application unit, accommodates at least part of a proximal end side of the conductive unit, and is inserted into the first hub; and a liquid circulation unit that allows the liquid supplied from the port unit to be sent to the distal end portion of the first shaft, wherein the energy application device is relatively movable forward and backward with respect to the catheter device, so that the energy application unit can be placed at a predetermined position on a distal end side of the first shaft, the liquid circulation unit is a fluid communication passage that includes the port unit of the second hub, an internal space of the second hub, a lumen of the second shaft, an internal space disposed immediately inside the first hub, and a lumen of the first shaft, wherein these components are in serial communication, and the liquid circulation unit includes a clearance around the electrical connector unit in the internal space of the second hub.

2. The medical device according to claim 1, wherein the second shaft is inserted into the first hub, to allow a distal end portion of the second shaft to move between a first position defined in the first hub and a second position defined closer to a proximal end side in the first hub than the first position, and the liquid circulation unit is provided in both a state where the distal end portion of the second shaft is located at the first position and a state where the distal end portion of the second shaft is located at the second position.

3. The medical device according to claim 1, wherein the one or more openings include: a first opening formed at a distal end of the first shaft; and a plurality of second openings that is located on a proximal end side of the first opening and is open toward a lateral side of the first shaft.

4. The medical device according to claim 3, wherein the plurality of second openings is open in a direction not to face an inner wall of the blood vessel in a state where the first shaft is inserted in the blood vessel.

5. The medical device according to claim 4, wherein the distal end portion of the first shaft

includes: a curved portion that has a shape curved with respect to a predetermined first reference axis; and a linear portion that is located on a proximal end side of the curved portion, and extends substantially linearly along a second reference axis extending in a direction intersecting with the first reference axis.

6. The medical device according to claim 5, wherein the curved portion is a spiral portion that extends spirally around the first reference axis.

7. The medical device according to claim 5, wherein the first shaft includes: a first portion included in the curved portion; a second portion that is located on a proximal end side of the first portion, wherein the plurality of second openings are formed in the second portion; and a third portion that is located on a proximal end side of the second portion, and is included in the linear portion, and an inner diameter of the second portion and the third portion is larger than an inner diameter of the first portion.

8. The medical device according to claim 1, wherein the energy application unit is capable of emitting a microwave or an ultrasonic wave as the energy.
